

EXPERIMENT-7

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Source Code:

```
main.py
1 import hmac
2 import hashlib
3
4 key = input("Enter secret key: ").encode()
5 message = input("Enter message: ").encode()
6
7 mac = hmac.new(key, message, hashlib.sha256).hexdigest()
8
9 print("Message Authentication Code (MAC):", mac)
10
11 message2 = input("Enter message to verify: ").encode()
12 mac2 = hmac.new(key, message2, hashlib.sha256).hexdigest()
13
14 if hmac.compare_digest(mac, mac2):
15     print("Message is authentic.")
16 else:
17     print("Message is not authentic.")
```

Output:

```
Output
Enter secret key: khushi
Enter message: hello
Message Authentication Code (MAC):
7ae10955025284569d2bf2a90349aec4cda12c11928ed25074e4f4d3e26a27dd
Enter message to verify: hello
Message is authentic.
```